Lesson 04 Demo 01

Demonstrating Basic Routing in a React Application

Objective: To demonstrate the implementation of client-side navigation by creating and managing routes in a React application using Vite and React Router

Tools required: Visual Studio Code and Node.js

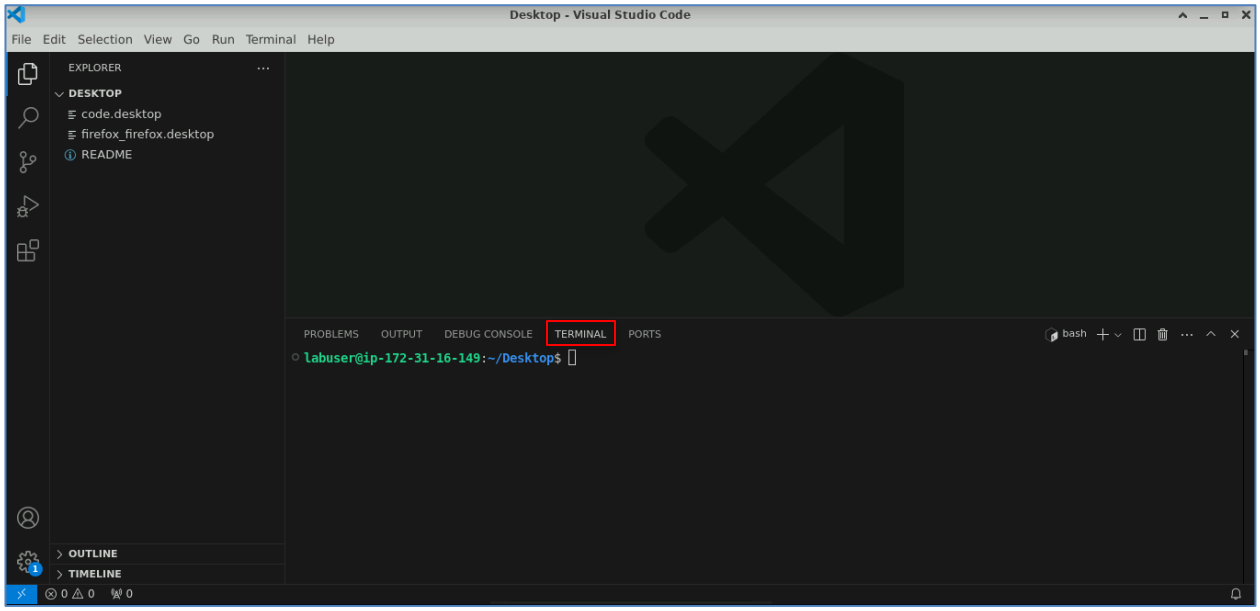
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create React components
3. Import and use the components
4. Run and verify the application

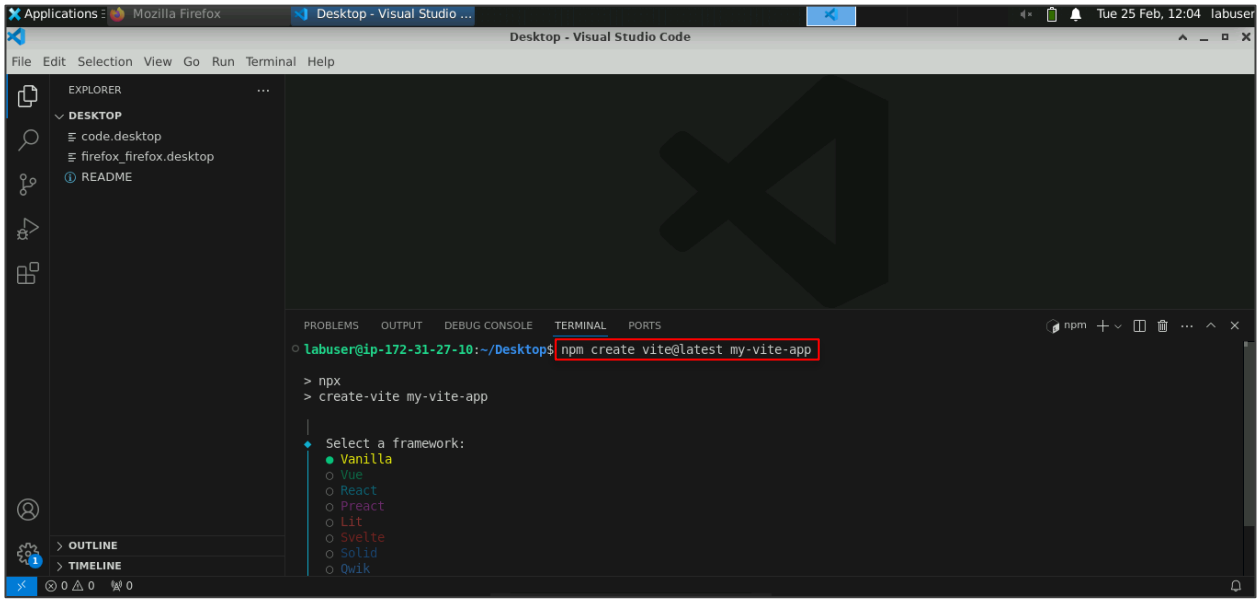
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

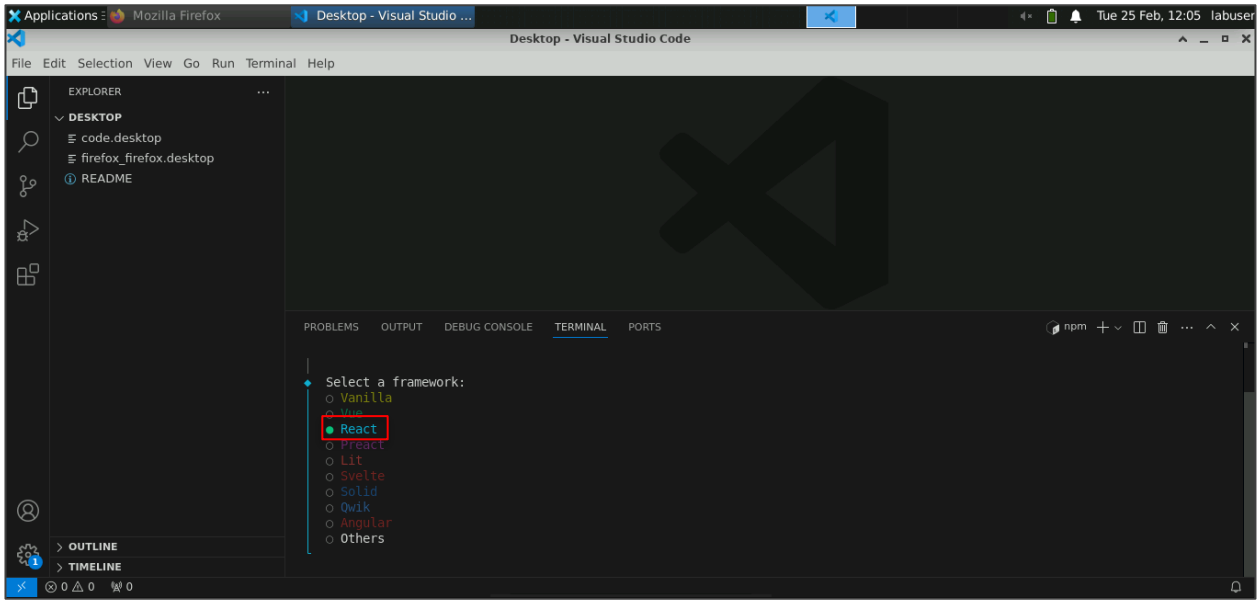


1.2 Run the following command to create a new React project using Vite:

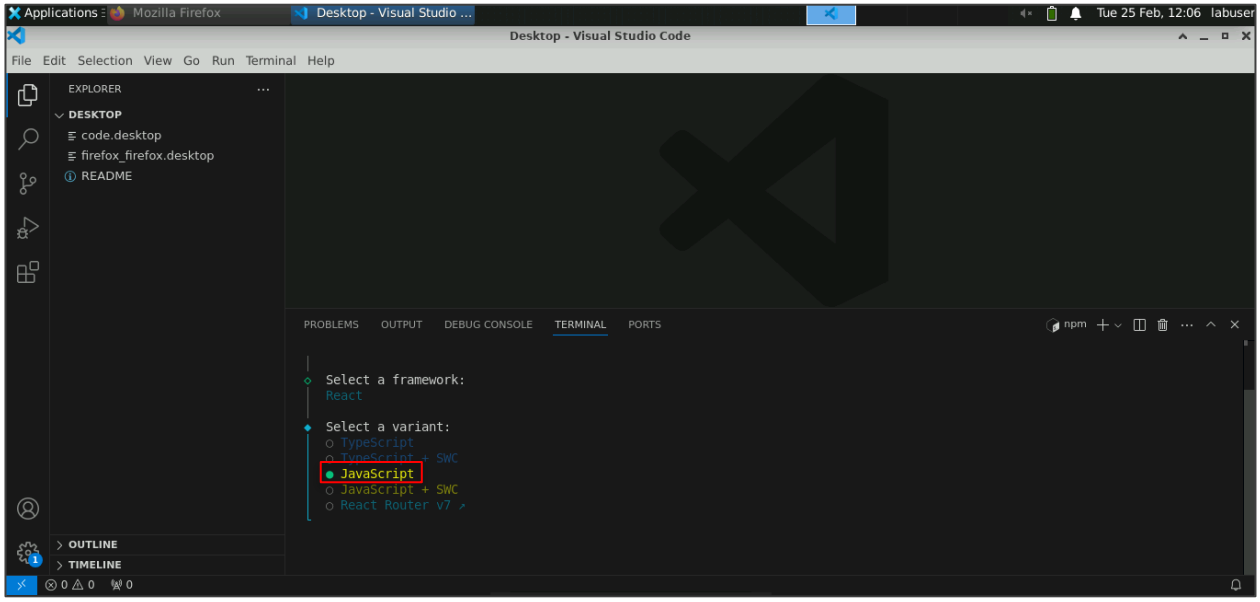
npm create vite@latest my-vite-app



1.3 Select **React** and press enter

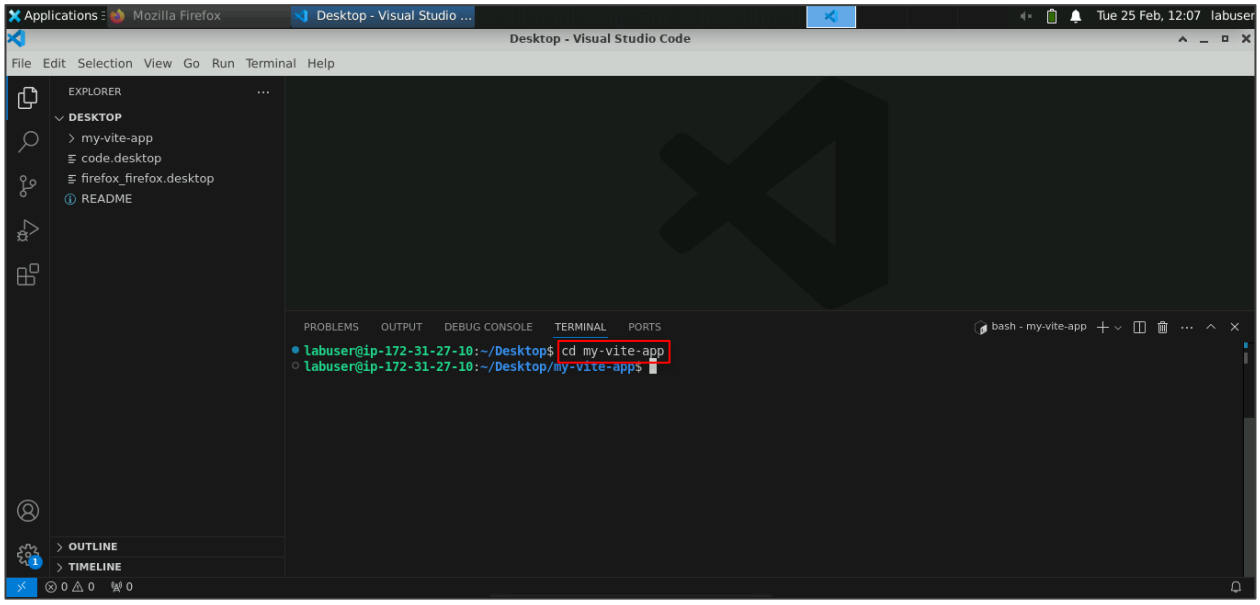


1.4 Select **JavaScript** and press enter



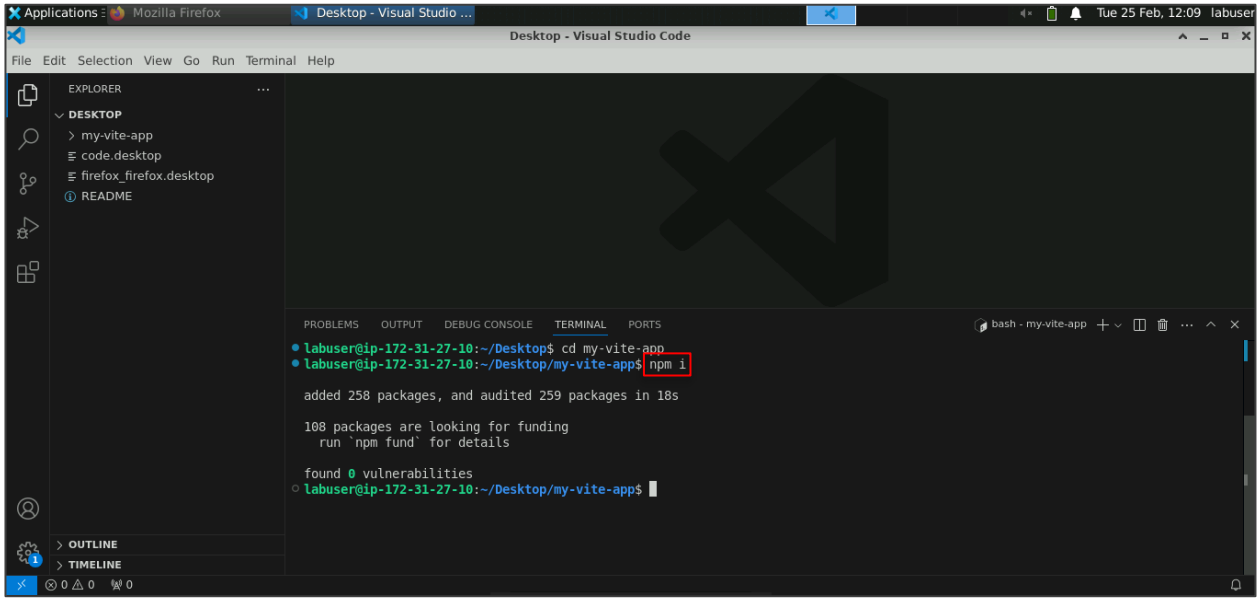
1.5 Run the following command to navigate to the project folder:

cd my-vite-app



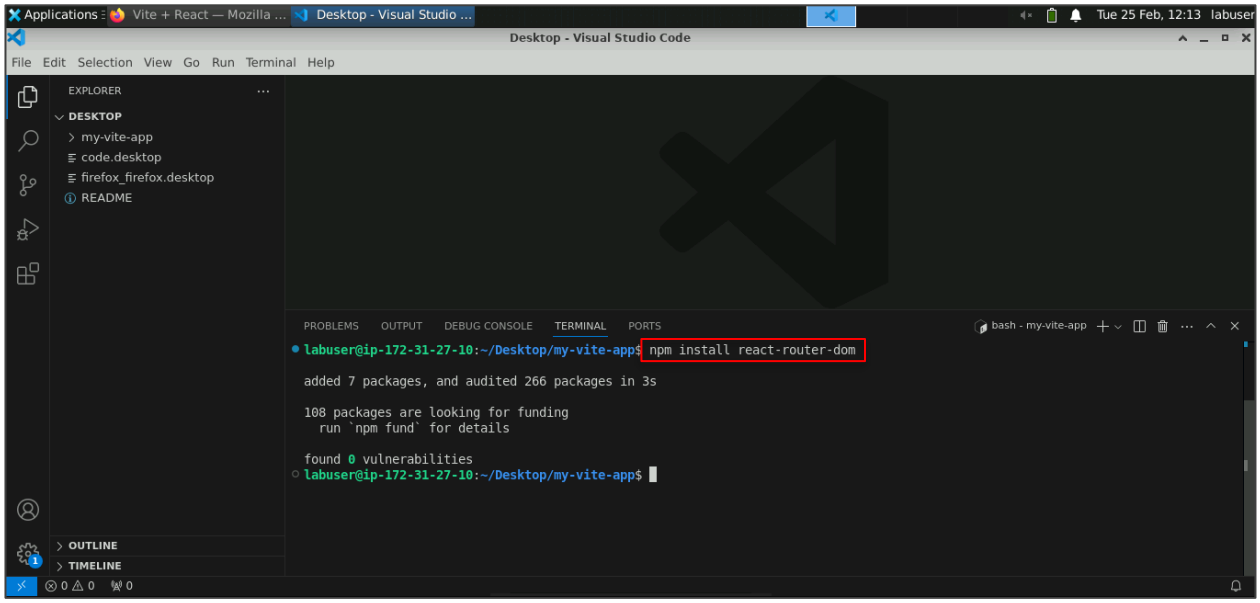
1.6 Run the following command to install dependencies:

npm i



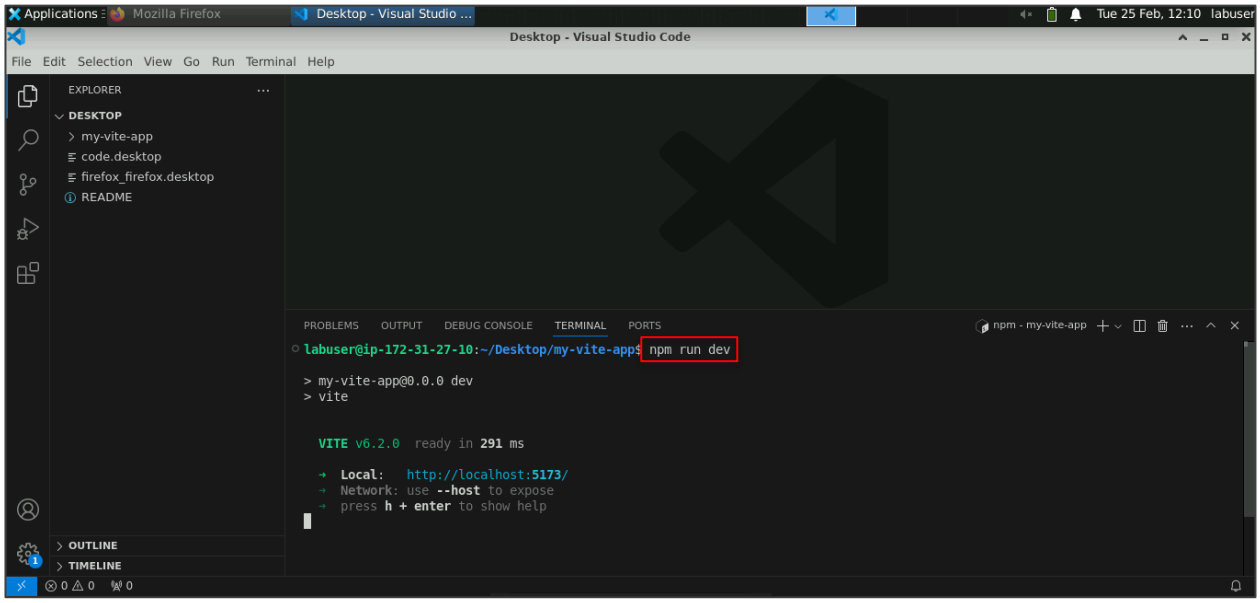
1.7 Run the following command to install the React router:

npm install react-router-dom



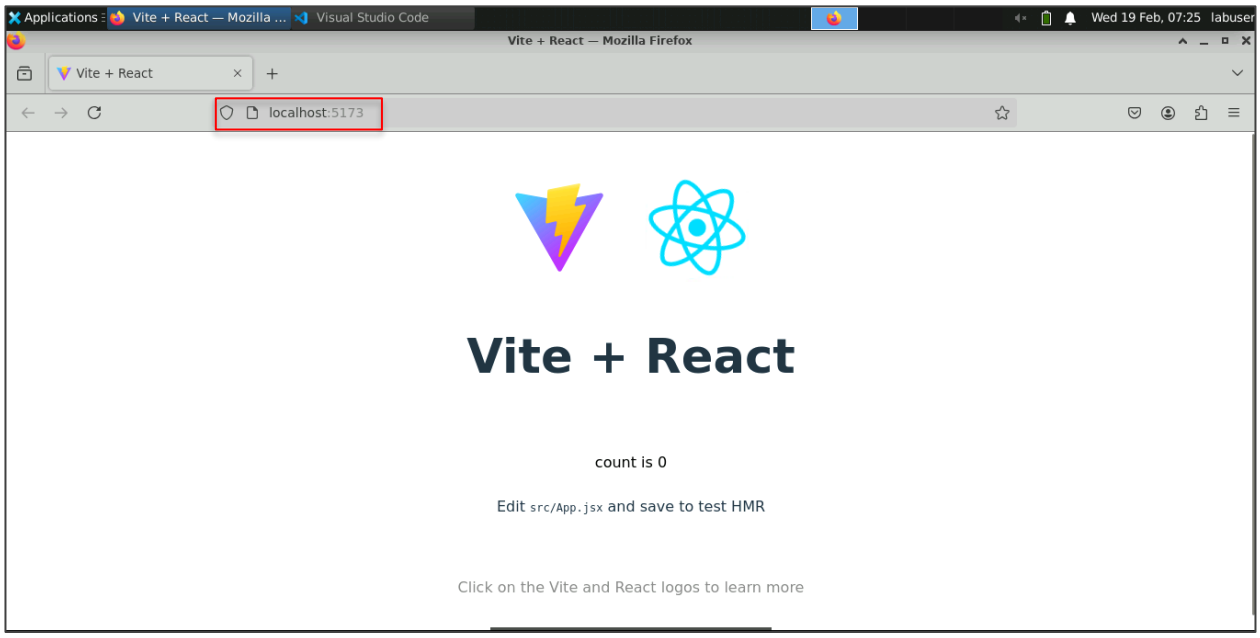
1.8 Run the following command to start the development server:

npm run dev



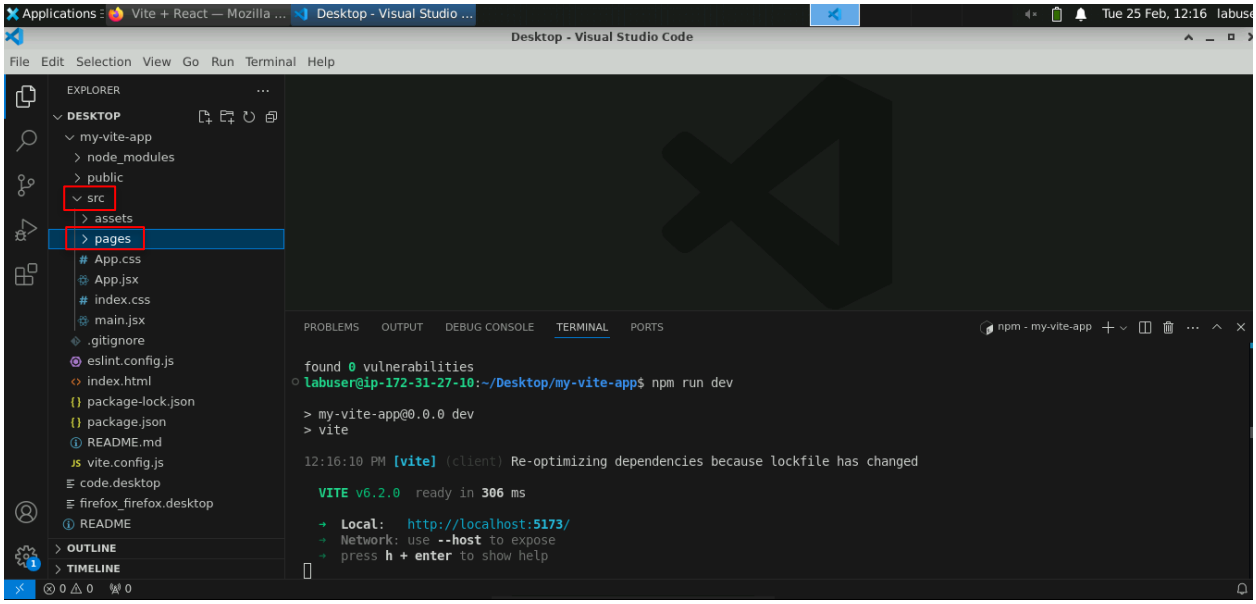
1.9 Navigate to the web browser and paste the link below:

http://localhost:5173

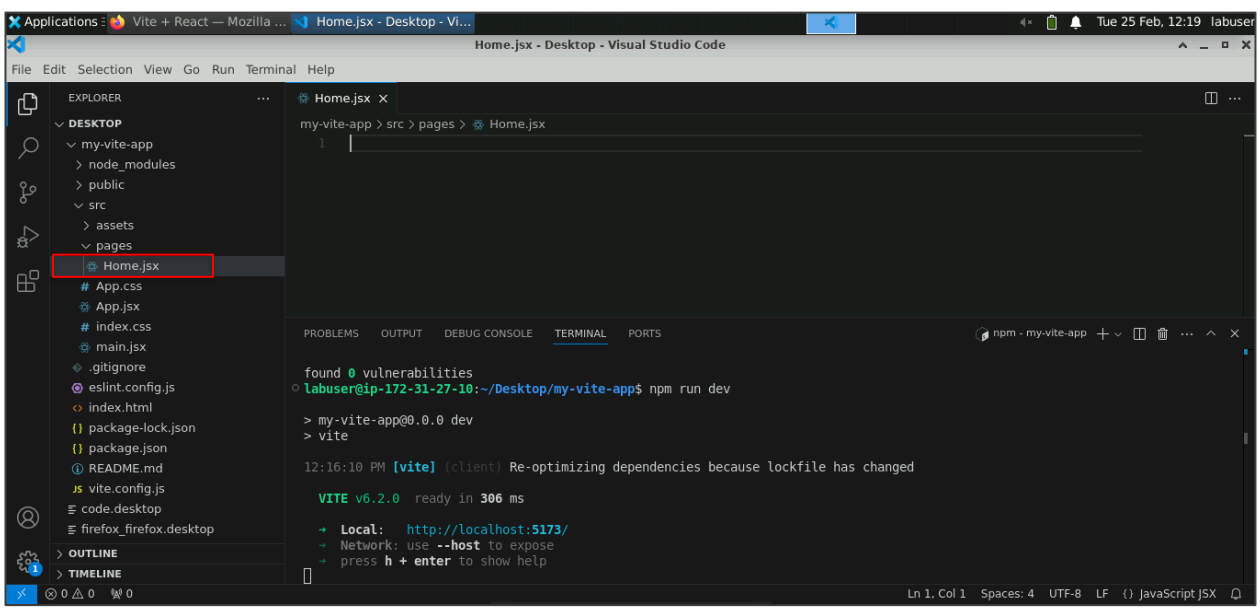


Step 2: Create React components

2.1 Inside the **src** folder, create a new folder named **pages**

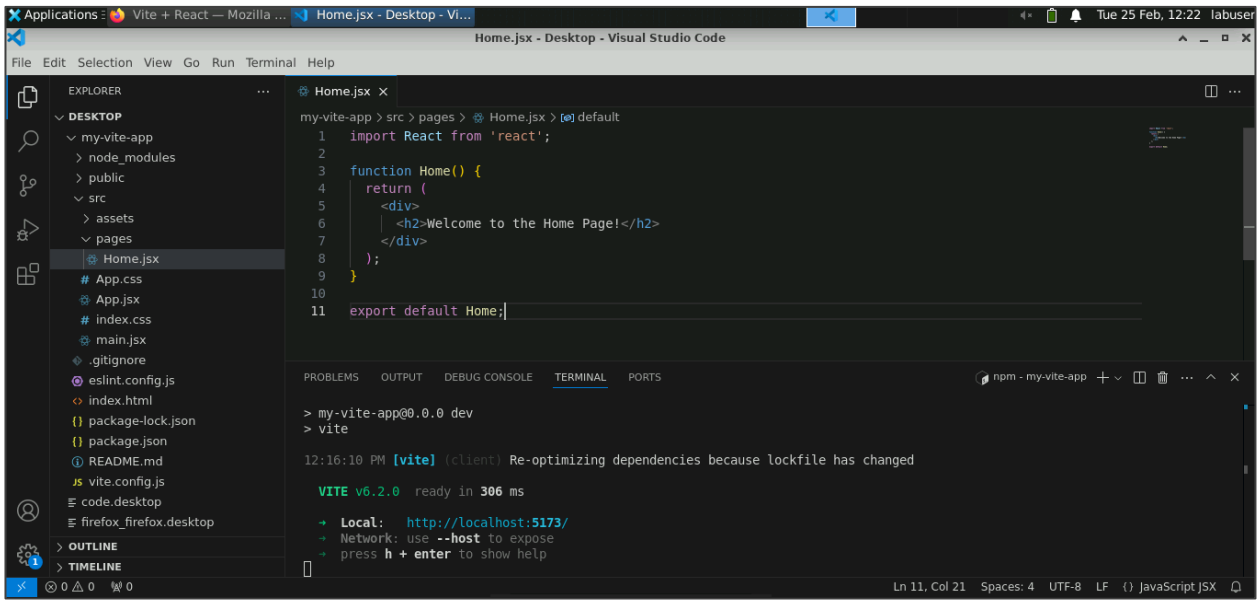


2.2 Inside the **pages** folder, create a new file named **Home.jsx**

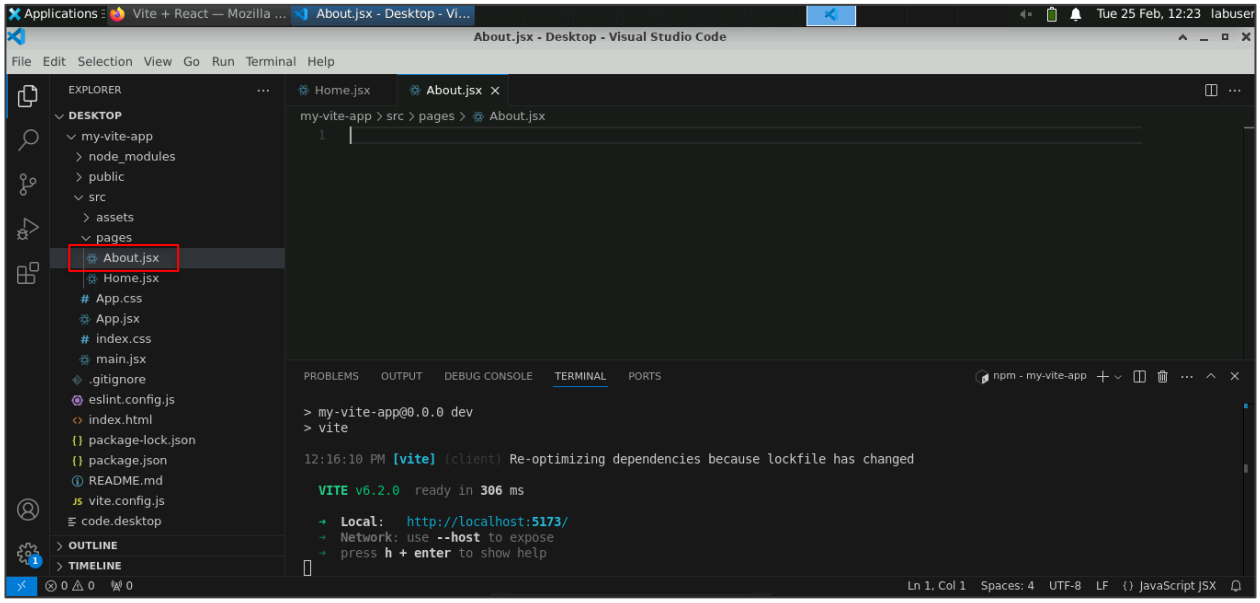


2.3 Add the following code to **Home.jsx:**

```
import React from 'react';

function Home() {
  return (
    <div>
      <h2>Welcome to the Home Page!</h2>
    </div>
  );
}

export default Home;
```

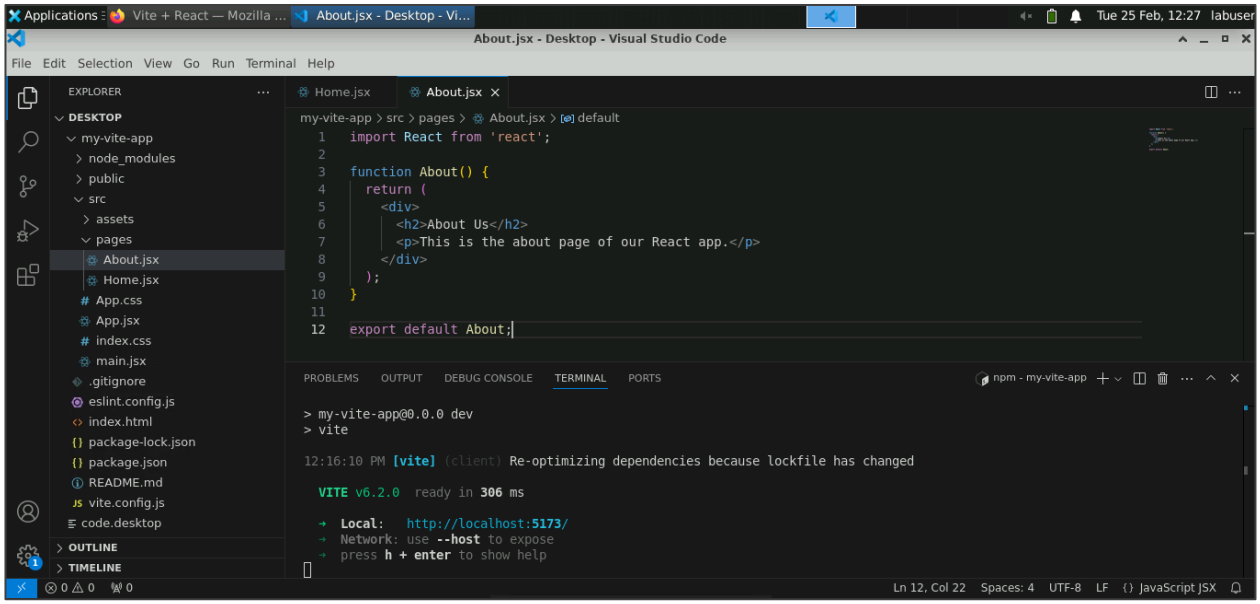


2.4 Inside the **pages** folder, create a new file named **About.jsx**



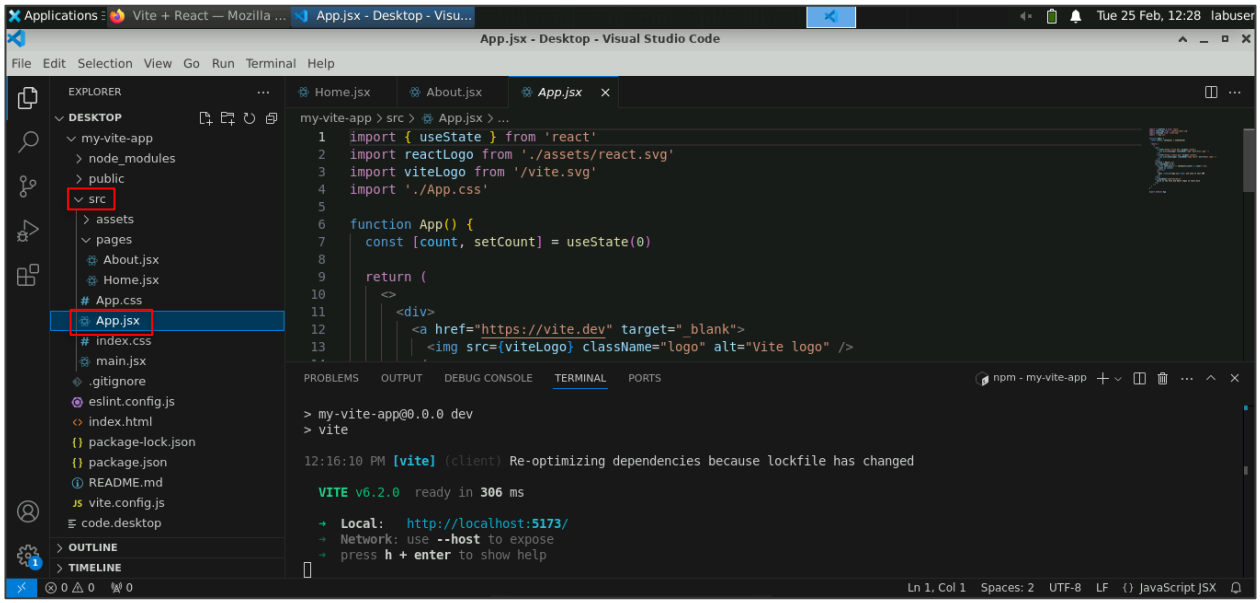
2.5 Add the following code to **About.jsx:**

```
import React from 'react';

function About() {
  return (
    <div>
      <h2>About Us</h2>
      <p>This is the about page of our React app.</p>
    </div>
  );
}

export default About;
```



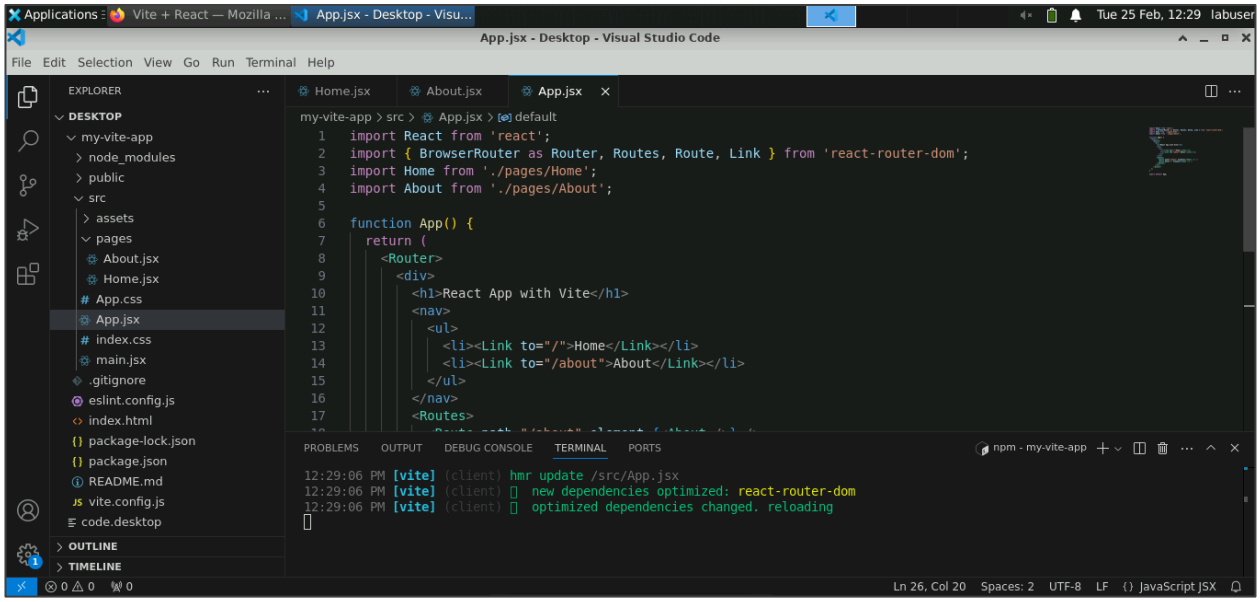
Step 3: Import and use the components

3.1 Open the **App.jsx** file inside the **src** folder



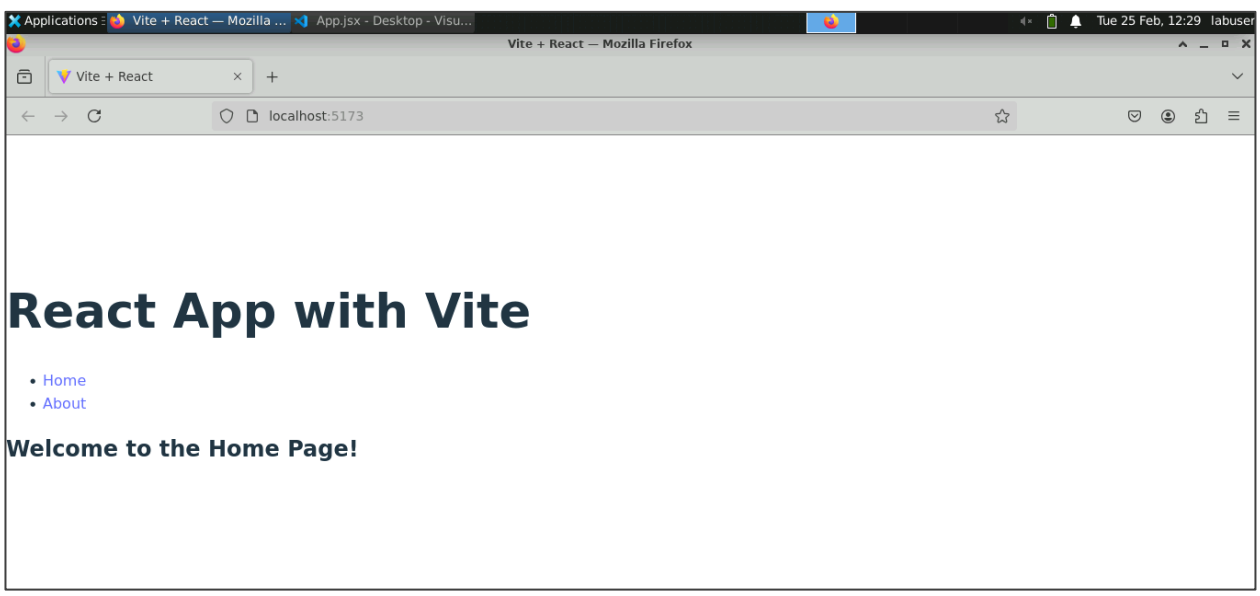
3.2 Paste the code below to include the new components in **App.jsx**:

```
import React from 'react';
import { BrowserRouter as Router, Routes, Route, Link } from 'react-router-dom';
import Home from './pages/Home';
import About from './pages/About';

function App() {
  return (
    <Router>
      <div>
        <h1>React App with Vite</h1>
        <nav>
          <ul>
            <li><Link to="/">Home</Link></li>
            <li><Link to="/about">About</Link></li>
          </ul>
        </nav>
        <Routes>
          <Route path="/about" element={<About />} />
          <Route path="/" element={<Home />} />
        </Routes>
      </div>
    </Router>
  );
}

export default App;
```

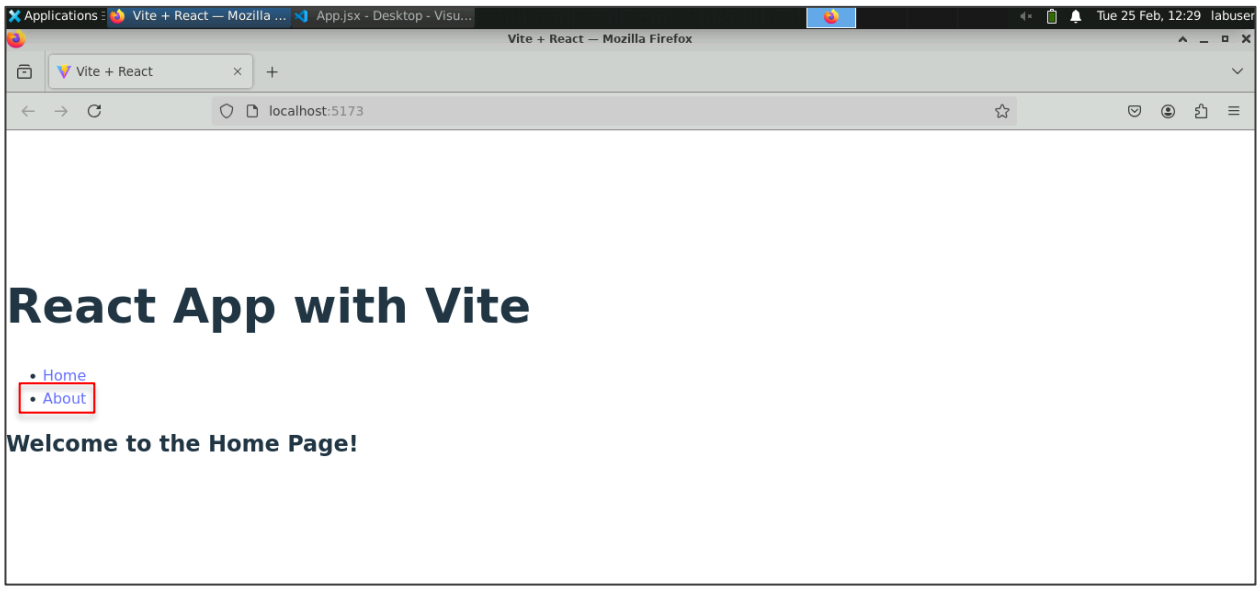


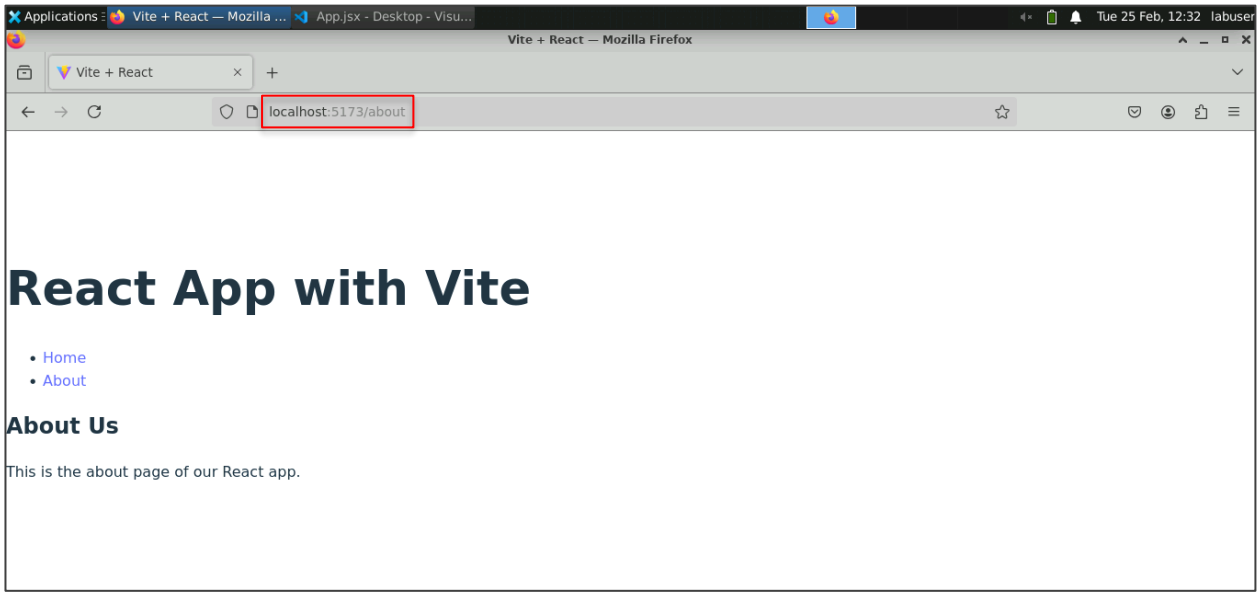
Step 4: Run and verify the application

4.1 Navigate to the browser and refresh the webpage



4.2 Click **About** to navigate to **/about** route





By following these steps, you have successfully implemented basic routing in a React application using React Router with Vite, enabling smooth navigation and rendering between different pages or components.